

The Shared Narrative of the ClimOO Ontology Suite

The following table provides the shared and participatory narrative that accounts for and accommodates perspectives, frameworks of reference, and purposes of involved stakeholders, concerning sustainability, plastic pollution, and climate change, especially in relation to agrifood systems. The ClimOO Ontology suite is constructed upon such a narrative.

Legenda:

XXXX = to further revise.

ClimOO Ontology Shared Narrative
Sustainability
Involved perspectives on sustainability: ● Sustainability interpreted through Hans Carl von Carlowitz's view: A balance of resources of the system of interest. Focus: Environmental performance assessment and analysis. Continuation of processes occurring in the system of interest. ● Sustainability as a commitment towards a sustainable development of the system of interest and the perpetual change characterizing such development, far from the system collapsing. Sustainable development as the permanent reestablishment of opportunities for system novelty and evolution that guarantee living conditions that are valuable and considered valuable by all system components, and among which they can opt freely (Ubbiali, 2025). ● Sustainability interpreted in terms of the “our common future”; the Brundtland Report view. ● Sustainability interpreted as an integration of environmental, social, and economic dimensions. The “three pillars” view. ● Sustainability in terms of the Environmental, Social, and Governance (ESG) principles. Focus on Environment and Society. Frame of reference: SDGs. ● Sustainability in terms of the propagation (at least not diminishing) of certain settled values for all humans. Sets of values of reference: SDGs. Main relevant stakeholders (especially with regards to agricultural systems): ● All living beings, including humans, related institutions, animals, microbes, and plants. Specific examples from human and related institutions’ sets: farmers, consultancy services, cooperatives,

farmers' associations, political decision-makers (regional government), researchers, marketers, and the local and regional communities, including:

- The South Tyrolean Extension Service for Fruit and Wine Growing
<https://www3.beratungsring.org/it/home>
- Consortium of South Tyrolean Fruit Growers' Cooperatives (VOG)
<https://www.vog.it/en/home.html>
- The Autonomous Province of Bolzano–South Tyrol (Italy), in particular, the “ [Incarico speciale complesso "Sostenibilità" office](#).

Main system scales of analysis:

- Geographical, spatial, social, and economic scales.

Main levels of analysis:

- Farm, region, national, and (international) levels.

Main systems of interest:

- The agricultural systems, specifically the South Tyrol region (Italy) apple production systems.

System dynamics and interactions:

- Focus on South Tyrolean agri-food system dynamics.
- Emphasis on interactions concerning the plastics lifecycle, related pollution, and interconnection with climate change.

System components:

- Farms, fields, inputs (seeds, fertilizers, pesticides), outputs (harvested products).

Activities, actors, and supply chains:

Farmers and related businesses (especially family and small businesses; average ~3 ha in fruit growing), food cooperatives, food retailers and related businesses, food distributors and related businesses, and food processors and related businesses.

Agriplastics:

mulch film

plastic-coated seed

biosolid material

plastic container

plastic silage wrap

greenhouse tunnel

Alternative to plastics

Transformations of agricultural systems should:

- Ensure farm survival and agricultural production while conserving system resources such as soil, water, and environment, especially with regard to businesses in small geographical areas.
- Foster a decrease in environmental contamination, especially plastic contamination.
- Promote biodiversity, habitat protection, and creation.
- Sustain adaptation to system dynamics, for instance, to variations in price pressure and regulatory requirements.
- Ensure effective management of land prices and production costs.
- Foster fair working conditions and salary.
- Incentivize marketing support, through, for instance, direct marketing, farm sales, and cooperation with tourism.
- Incentivize local food consumption by local communities.
- Safeguard farm succession and farm internal social structure (e.g. caring for older generations).
- Promote the usage of reusable, biodegradable, or recyclable plastics.
- Point to reduce the usage of plastics and implement viable and sustainable alternatives.
- Discourage plastic accumulation of plastics and plastic debris in agricultural soil and water systems.
- Incentivize the usage of energy from renewable sources and reduce raw material usage.
- Promote the reduction of greenhouse gas emissions.
- Leverage on resource-saving cultivation and climate-smart methods.
- Foster the cultivation of climate-change resilient varieties of plants.
- Support production practices that safeguard resource recovery and benefits beyond solely human benefits, e.g. reducing the usage of pesticides.
- Establish clear and shared policies and standards for plastic regulation and definition. To assess and implement existing references.

To implement through forward-looking strategies, especially strategies that promote a circular economy.

Primary issues to target:

- No circularity in the life cycle of used plastic.
 - Environmental impacts of short-term single-use plastics.
 - Few alternatives, generally too expensive.
- Contamination of soil and water.
 - Soil and water quality.
 - Alteration of soil structure.
 - Variation in soil water retention capacity.
 - Alteration of the soil organic matter.
- Depletion of system resources.
- Greenhouse gas emissions from agricultural supply chains.
- Extreme weather events.

Features to targets:

System resilience, system stability, system adaptability, human well-being and animal welfare, food safety, food locality, economic viability, biodiversity, living and working conditions, and farm and local community social structure

Plastics and (micro-)plastic pollution**Plastics and plastic pollution:**

- To date, there has been an overuse of traditional plastics due to their versatility and usefulness across fields and domains. No material can currently substitute plastics adequately.
- Plastics and plastic debris are present everywhere, including oceans and water streams, soils, and human bodies.
- There exists a distinction among plastic polymers, plastic materials, and plastic objects/particles, although the terminology is sometimes misused interchangeably.
 - Polymers, materials, and pieces of plastic can be mainly categorized as virgin, recycled, recyclable, biobased, biodegradable, and compostable.
 - Synthetic polymers, materials, and related items and objects are outputs of some human manufacturing production process.
- Synthetic plastics are mainly composed of synthetic plastic polymers, constituting the material matrix, mixed primarily with additives. Sometimes the percentage of additives can exceed 50% of the total mass, but generally it is the previous consideration that applies (Hartmann et al., 2019; Masterton et al., 2023).
 - Examples: PE, PVC, PLA, PA, and PP.

- If released into the soil and water systems, additives become contaminants, affecting plant growth, soil microbiota, and food safety. There is the possibility of interaction among different pollutants.
 - Main types of additives: plasticizers, stabilizers, flame retardants, flow modifiers, processing aids, impact modifiers, antioxidants, pigments, biocides, and fragrances (Masterton et al., 2023).
- Main features to deal with: biodegradability, compostability, biobaseness, recyclability, plastic polymer degradation, plastic fragmentation, and environmental persistence of plastic debris. To pay attention to possible misinterpretations and shortcomings of current regulations.
 - Biobased and biodegradable plastics are often seen as alternatives to traditional plastics. Nevertheless, several works raised concerns regarding deriving issues. See, for instance, the works of Carteny et al. (2022) and Steiner et al. (2022, 2024).
- Properties such as bio-baseness and biodegradability primarily apply to polymers. Example from Summary 2: there is an increased usage of such materials and objects in agriculture. Nevertheless, despite being produced using biodegradable polymers, they are not (at least completely or they are only under certain conditions/in certain contexts) biodegradable, due to their structure or chemical composition, e.g. presence of additives. Biodegradability and biobaseness are distinct properties, and a biobased plastic material is not necessarily biodegradable. Regulations and standards play a fundamental role in defining the threshold of applicability and measurement of such properties (See for instance Rayns et al., 2021; Steiner et al., 2022).
- Plastic debris are pieces of synthetic polymer-based materials that do not fulfill any purposes, are environmental pollutants, are solid and insoluble in water, and are mobile and persistent in environmental systems. See also the work of Hartmann et al. (2019).
 - According to Hartmann et al. (2019), plastic debris does not refer solely to entities made of plastic materials, but to pieces, more generally, made of synthetic polymer-based materials (Hartmann et al. 2019).
- Different methods for analysis and classification of plastic debris exist (Bank et al., 2022; Hartmann et al., 2019; Rochman et al., 2019; Yuan et al., 2022). Main criteria for classifying plastic debris include chemical composition, solid state, solubility, size, shape and structure, color, and origin. (Hartmann et al., 2019; Rochman et al., 2019).
 - Based on the size criterion: megaplastic particle, macroplastic particle, mesoplastic particle, microplastic particle, and nanoplastic particle.
 - No common agreement exists on size ranges for classification.

- Microplastics are generally categorized as particles smaller than 5 mm in their largest dimension. Alternative systems of classification set the range between 1 and 1000 µm in its largest dimension (Hartmann et al. 2019).

Plastics in agricultural systems:

- Different usages depending on cultural and legal aspects, types of crops, and cultivation practices.
- Main usage: crop protection, soil and microclimate management, irrigation, plant support and binding, cultivation input management, packaging and transportation (Rayns et al., 2021; UNEP, 2022).
- Short-term single-use and long-term use plastic.

Plastics in South Tyrolean agrifood systems:

- Plastics are used primarily for plant binding.
- There is a limited use of plastic films in apple orchards. An exception is the usage of fruit coloring films. But there is no standardized usage.
- There is a limited use of compost.
- Packaging materials: out of our system of interest.

Plastic pollution:

- During plastic pollution processes, plastic debris are released into and possibly accumulated in an environment, from some plastic debris source to some plastic debris sink, affecting the structure and properties of the environment itself and those of the related system components (Du et al., 2021; Eerkes-Medrano & Thompson, 2018; El-masry et al., 2025; Liro & Zielonka, 2025; Su et al., 2022).
 - Plastic debris degradation and fragmentation, release and sorption of carried pollutants may occur in parallel (Fu et al., 2021; Liro and Zielonka, 2025; Masterton et al., 2023).
- Plastic persistence and fragmentation in soils affect the soil structure and porosity, water retention, aeration, and microorganisms, as well as humus formation, availability of water, and nutrients (Eerkes-Medrano and Thompson, 2018; Gerassimidou and Iacovidou, 2024).
- Plastics also contaminate groundwaters and surface waters (Du et al., 2021; Eerkes-Medrano & Thompson, 2018).

Contributions to climate change:

- Plastic production and disposal contribute to greenhouse gas emissions and resource consumption and depletion.

- The usage of plastics during agricultural activities contributes to climate change less than these other activities.

- Accumulation of microplastics in the soil affects, among others, the soil structure, contributing to greenhouse gas emissions and impacting soil carbon storage capacity (Chang et al., 2024; Ujuagu et al., 2025; Wang et al., 2025).

Types of plastics and extent in food systems, sources, and sinks.

Plastic threats in agriculture.

Alternative to plastics.

Climate change (especially in agricultural systems)

Main dimensions of relevance:

- Climatic conditions.
- Consequences on agricultural systems.
- Physiological responses of plants.

There is a need to cover the implications of climate change for humans and beyond, e.g. for animals and the environment.

Climate change in agriculture:

- Average temperature variation increases the risk of frost damage (late frost, early budburst or flowering), emergence of invasive pests, outbreak of novel diseases, and variation of pest and disease pressure.
- Heat damage and stress.
- Drought stress.
- Decrease in food quality (fruits) and storability.
- Alteration of the development of fruit aroma, color, and formation, and an increase in the rate of physiological disorders.
- Increase in production costs.
- Increase in extreme (climate) events.
- Variation of production conditions.
- Increase in uncertainties (in yields, quality, and management strategies).

Contributions to plastic pollution:

- An increase in temperature and UV radiation contributes to plastic debris fragmentation and soil and water pollution.

- An increase in the occurrence of storms and floods contributes to plastic debris fragmentation and soil and water pollution.

Contributions to plastic pollution:

Main frameworks of reference on sustainability, plastic pollution, and climate change

EU and Italy policies

- EU Commission Press Release ip_23_4581. https://ec.europa.eu/commission/presscorner/detail/en/ip_23_4581
- EU Common Agriculture Policy (CAP). https://agriculture.ec.europa.eu/common-agricultural-policy/cap-overview/cap-glance_en
- EU Corporate Sustainability Reporting Directive (CSRD). https://finance.ec.europa.eu/capital-markets-union-and-financial-markets/company-reporting-and-auditing/company-reporting/corporate-sustainability-reporting_en
- EU EN 13432 bioplastic policy. <https://www.en-standard.eu/bs-en-13432-2000-packaging.-requirements-for-packaging-recoverable-through-composting-and-biodegradation.-test-scheme-and-evaluation-criteria-for-the-final-acceptance-of-packaging/>
- EU organic farming policy. https://agriculture.ec.europa.eu/farming/organic-farming/organics-glance_en
- EU Waste Codes:
 - EU Waste Directives
 - https://environment.ec.europa.eu/topics/waste-and-recycling/implementation-waste-framework-directive_en
 - European Wastes Catalogue (EWC) <https://www.eea.europa.eu/help/glossary/eea-glossary/european-waste-catalogue-1>

Further standards

- EUR - Lex document 32023R2055. <https://eur-lex.europa.eu/eli/reg/2023/2055/oj/eng>
- UNI CEN/TS 15968:2010 - Determination of extractable perfluorooctanesulphonate (PFOS) in coated and impregnated solid articles, liquids and firefighting foams - Method for sampling, extraction and analysis by LC-qMS or LC-tandem/MS. <https://www.en-standard.eu/csn-p-cen-ts-15968-determination-of-extractable-perfluorooctanesulphonate-pfos-in-coated-and->

[impregnated-solid-articles-liquids-and-fire-fighting-foams-method-for-sampling-extraction-and-analysis-by-lc-qms-or-lc-tandem-ms/](#)

- UNI EN ISO 20200:2023 - Plastics — Determination of the degree of disintegration of plastic materials under composting conditions in a laboratory-scale test.
<https://www.iso.org/standard/81932.html>
- UNI EN ISO 24187:2023 - Principles for the analysis of microplastics present in the environment.
<https://www.iso.org/standard/78033.html>
- UNI ISO 4484-2:2023 - Textiles and textile products — Microplastics from textile sources; Part 2: Qualitative and quantitative analysis of microplastics. <https://www.iso.org/standard/80011.html>
- ISO 14030-3:2022 - Environmental performance evaluation — Green debt instruments; Part 3: Taxonomy. <https://www.iso.org/standard/75559.html>

Regional guidelines

- AGRIOS guidelines. Gruppo di lavoro per la frutticoltura integrata. <https://www.agrios.it/it/>
- Bioland Südtirol guidelines <https://www.bioinsuedtirol.it/>
- Demeter guidelines. Demeter. <https://www.demeter.it/>
- L'Agenda 2030 del Consorzio Vini Alto Adige. Centro di competenza per il vino dell'Alto Adige.
<https://www.vinialtoadige.com/it/home/1-0.html>
- The South Tyrolean apple sector's sustainable development strategy:
 - The ABCs of sustainability. <https://www.sustainapple.it/en/>

Sustainapple - VIP Coop. soc. Agricola.

<https://www.vip.coop/en/business/sustainability/sustainapple/159-0.html>

Literature

- The works of Gold (2019), Hartmann et al. (2019), Kershaw & Galgani (2019), Rochman et al. (2019), and Vert et al. (2012).

Main Projects of reference

<https://www.research-alliance-for-sustainability.it/projects/>

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